Review

- **Origins of quantum theory**
- **Classical vs. Quantum**
- **The Schrödinger equation**

Postulate 1. The state of a system is described by a function **Ψ** of the coordinates and the time. This function, called the state function or **wave function**, contains all the information that can be determined about the system.

Postulate 6. The time development of the state of an undisturbed quantum-mechanical system is given by the Schrödinger time-dependent equation.

For a one-particle, 1 D system:

The time-independent Schrödinger equation

A special 1-D case, $V(x,t) = V(x)$

Stationary state (定态):

Probability density does not change with time.

Review

Operators carry out a mathematical operation on a function; transform the function into another function.

Postulate 2. To every physically observable property there corresponds a **linear Hermitian operator**. To find this operator, write down the classical-mechanical expression for the observable in terms of Cartesian coordinates and corresponding linear-momentum components, and then replace each coordinate x by the operator $\hat{x}$ and each momentum component p_x by the operator $\hat{p}_x$

Hamiltonian Operator

Operator Algebra

Commutator

Commute

Review

Postulate 2. To every physically observable property there corresponds a **linear Hermitian operator**.

Linear Operator:

Hermitian (厄米) Operator is a **linear operator** that satisfies

Or

for all well-behaved functions

Eigenfunction 本征函数, Eigenvalue 本征值

$$\hat{H}\psi = E\psi$$

Eigenvalue equation : 本征值方程

$$\hat{\Omega}\psi = \omega\psi$$

Eigenfunction ψ
Eigenvalue ω

(Operator)(Function)=(Constant Factor)(Same Function)

Example

$$(\frac{d}{dx})e^{2x} = 2e^{2x}$$

If $f(x)$ is an eigenfunction of a linear operator $\hat{\Omega}$,

$Cf(x)$ is an eigenfunction of $\hat{\Omega}$ with the same eigenvalue.

$$\hat{\Omega}f(x) = \omega f(x)$$

$$\hat{\Omega}[Cf(x)] = C\hat{\Omega}f(x) = C\omega f(x) = \omega[Cf(x)]$$

Two properties of the Hermitian operators:

1) Eigenvalues are real numbers;
2) Eigenfunctions with different eigenvalues are orthogonal.

Orthogonality: $\int \psi_i^{\star} \psi_j \, d\tau = 0$

Justification: The reality of eigenvalues ⟶ $\omega^* = \omega$

For a wavefunction ψ that is normalized and is an eigenfunction of an hermitian operator $\hat{\Omega}$ with eigenvalue ω, we can write

$$\int \psi^{\star} \hat{\Omega} \psi \, d\tau = \int \psi^{\star} \omega \psi \, d\tau = \omega \int \psi^{\star} \psi \, d\tau = \omega$$

However, by taking the complex conjugate we can write

$$\omega^{\star} = \left\{ \int \psi^{\star} \hat{\Omega} \psi \, d\tau \right\}^{\star} = \int \psi^{\star} \hat{\Omega} \psi \, d\tau = \omega$$

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1$$ **Normalized！归一的！**

Two properties of the Hermitian operators:

1) Eigenvalues are real numbers;
2) Eigenfunctions with different eigenvalues are orthogonal.

$$\text{Orthogonality: } \int \psi_i^{\star} \psi_j \, d\tau = 0$$

e.g. sin x and sin $2x$ are eigenfunctions of the Hermitian operator d^2/dx^2

$$\int \sin ax \sin bx \, dx = \frac{\sin(a-b)x}{2(a-b)} - \frac{\sin(a+b)x}{2(a+b)} + \text{constant}, \qquad \text{if } a^2 \neq b^2$$

It follows that, for $a = 1$ and $b = 2$, and given the fact that $\sin 0 = 0$, $\sin 2\pi = 0$, and $\sin 6\pi = 0$,

$$\int_0^{2\pi} (\sin x)(\sin 2x) dx = 0$$

Orthonormal
(正交归一)

$$\int g_i^{*} g_k \, d\tau = \delta_{ik}$$

$$\delta_{ik} = 1 \qquad i = k$$

$$\delta_{ik} = 0 \qquad i \neq k$$

Open Question Points: 20

Setting

Identify which of the following functions are eigenfunctions of the operator $\frac{d^2}{dx^2}$, and give the corresponding eigenvalue.

(a) e^x
(b) sinx
(c) 2cosx
(d) x^3
(e) sinx + cosx

Answer

Two minutes break!



Chapter 1

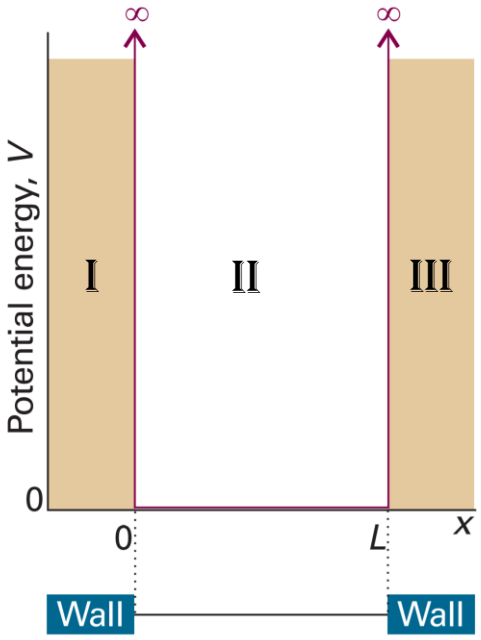
Quantum Theory

Part II

Contents

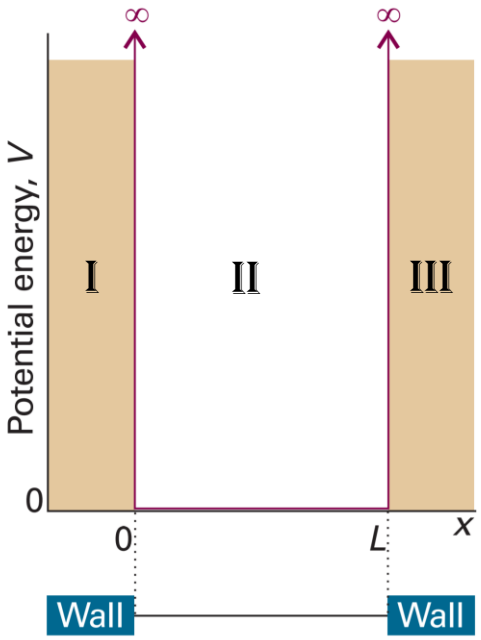
- A particle in a 1-D box
- The 1-D Harmonic Oscillator
- Measurement of a QM system

A particle in a 1-D box



$$V(x) = \infty \,, \quad x \in \mathrm{I}, \mathrm{III}$$
$$V(x) = 0 \,, \quad x \in \mathrm{II}$$

A particle in a 1-D box



$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$

For regions I and III

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}=(E-\infty)\psi$$

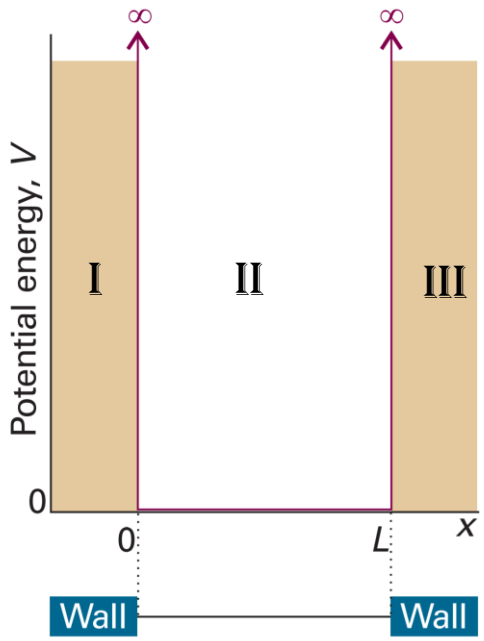
$$\frac{d^2\psi}{dx^2}=\infty\psi \quad\rightarrow\quad \psi=\frac{1}{\infty}\frac{d^2\psi}{dx^2}$$

$$\rightarrow\quad \psi(I,III)=0$$

$V(x)=\infty\ ,\quad x\in \mathrm{I},\ \mathrm{III}$

$V(x)=0\ \ ,\quad x\in \mathrm{II}$

A particle in a 1-D box



For region II

$$V(x) = 0 \qquad -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

General solutions:

$$\psi_k = Ae^{ikx} + Be^{-ikx}$$
$$= C\sin kx + D\cos kx$$

$$E_k = \frac{k^2\hbar^2}{2m}$$

Boundary conditions:

$$\psi_k(0) = 0, \quad \Rightarrow \quad \boldsymbol{D = 0} \qquad \boldsymbol{C \neq 0}$$

$$\psi_k(L) = 0, \quad \Rightarrow \quad \sin kL = 0$$

$$\Rightarrow \quad kL = n\pi, \quad n = 1, 2, \ldots$$

$$\psi_n(x) = \left(\frac{2}{L}\right)^{1/2}\sin(n\pi x/L) \qquad E_n = \frac{n^2h^2}{8mL^2} \qquad n = 1, 2, \ldots$$

***C*, obtain by normalization**

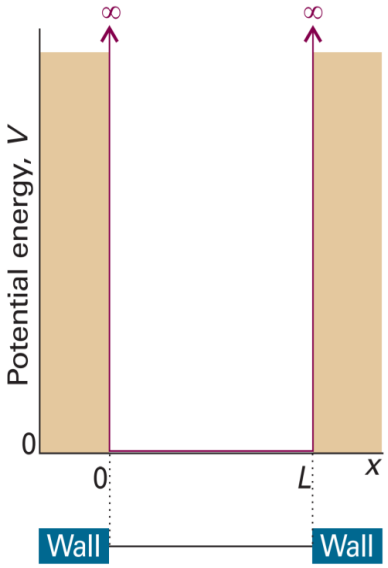
Quantized!

A particle in a 1-D box

$0 < x < L$

$$\psi_n(x) = \left(\frac{2}{L}\right)^{1/2} \sin(n\pi x/L) \qquad E_n = \frac{n^2h^2}{8mL^2} \qquad n = 1, 2, \ldots$$

Quantum Number



The lowest energy

$$E_1 = \frac{h^2}{8mL^2}$$ → **"Ground state"**

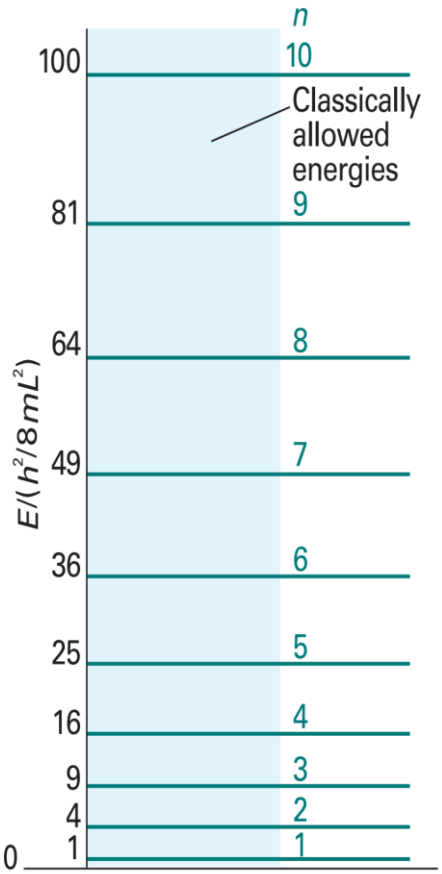
$n > 1$ → **"Excited state"**

Separation of adjacent levels

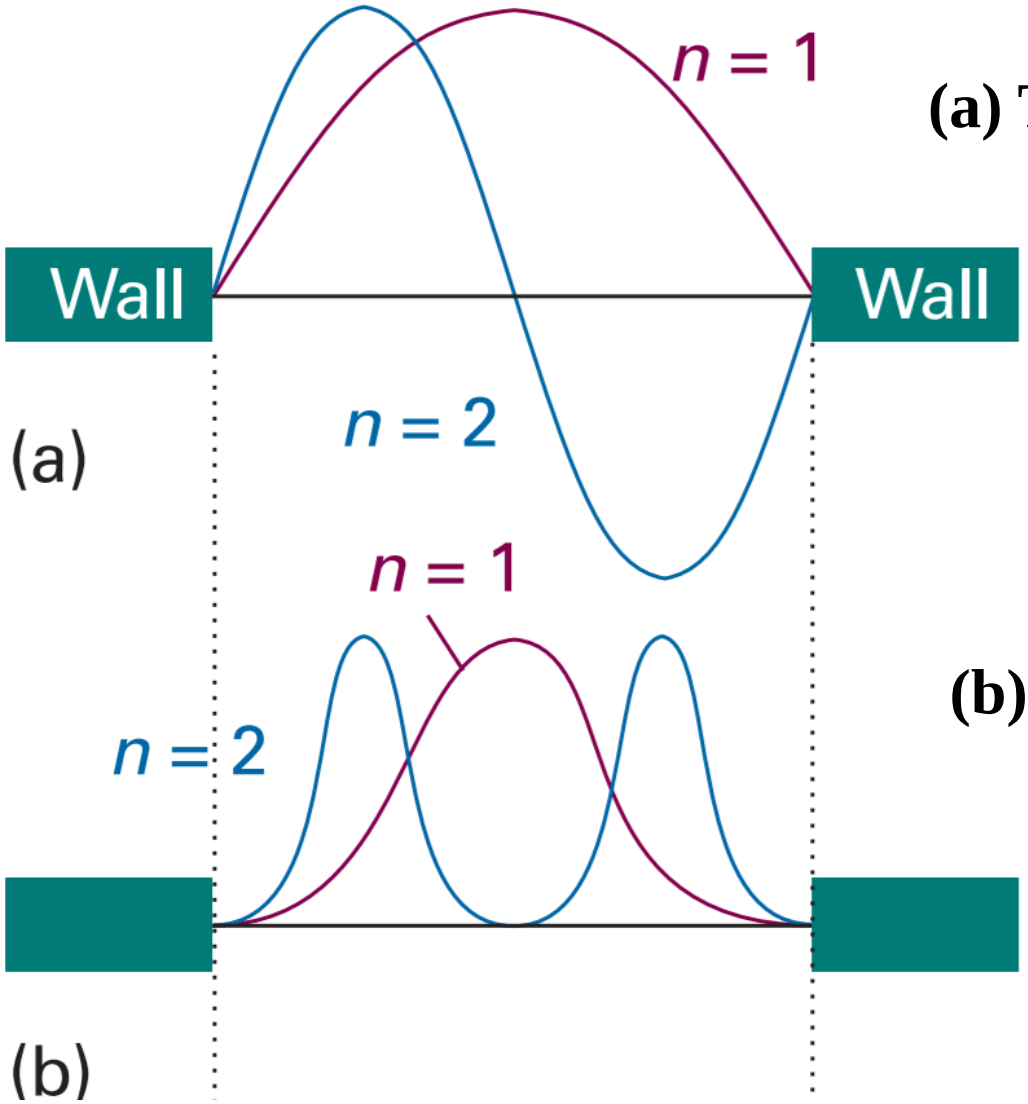
$$E_{n+1} - E_n = \frac{(n+1)^2h^2}{8mL^2} - \frac{n^2h^2}{8mL^2} = (2n+1)\frac{h^2}{8mL^2}$$

$x \to \infty, -\infty \qquad \psi \to 0$ **"Bound state"**

$x \to \infty, -\infty \qquad \psi \neq 0$ **"Unbound state"**



A particle in a 1-D box

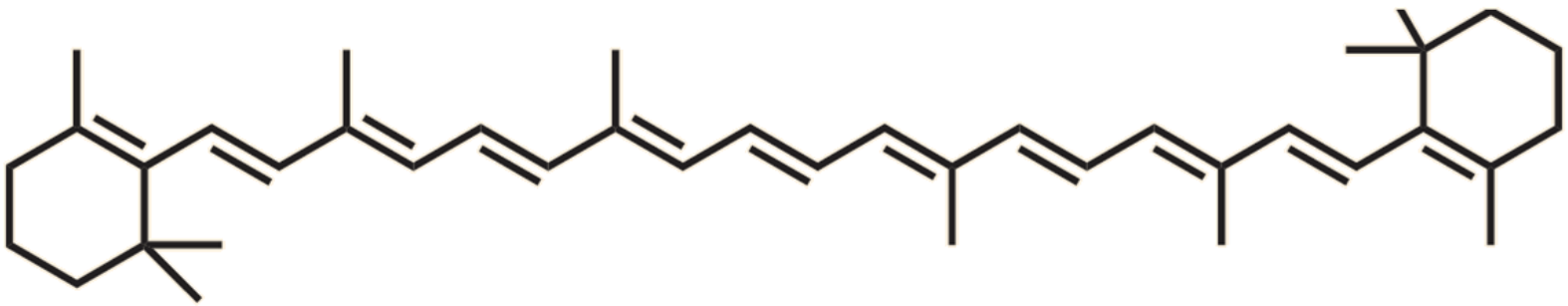


(a) The first 2 wavefunctions

(b) Probability density:

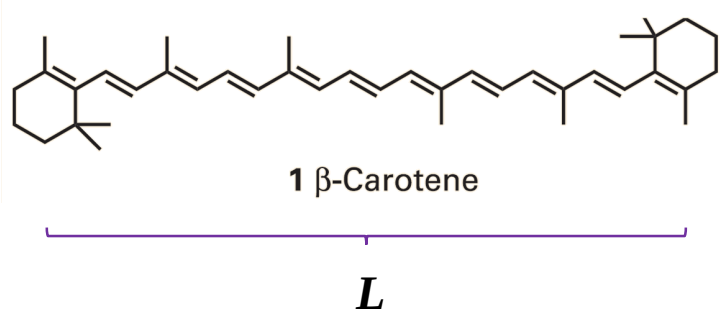
$$\psi^2(x) = \frac{2}{L}\sin^2(n\pi x/L)$$

e.g.



1 β-Carotene

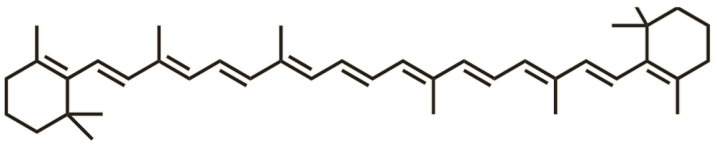
Take each CC bond as 140 pm.
What is the first excitation energy of the system?



Take each CC bond as 140 pm.
What is the first excitation energy of the system?

10 single and 11 double bonds, so L = 21*140 pm = 0.294 nm

Each of **22** conjugated carbons contributes one *p*-electron to π orbitals



1 β-Carotene

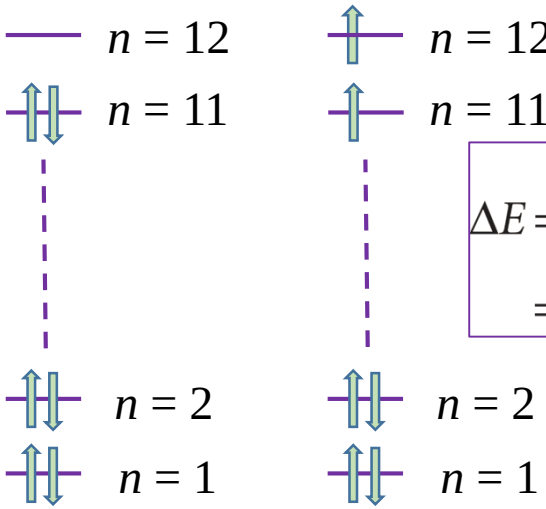
L

Take each CC bond as 140 pm.
What is the first excitation energy of the system?

10 single and 11 double bonds, so L = 21*140 pm = 0.294 nm

Each of **22** conjugated carbons contributes one *p*-electron to π orbitals

Ground state **1st exited state**



$$\Delta E = E_{12} - E_{11} = (2 \times 11 + 1)\frac{(6.626 \times 10^{-34}\ \text{J s})^2}{8 \times (9.110 \times 10^{-31}\ \text{kg}) \times (2.94 \times 10^{-10}\ \text{m})^2} = 1.60 \times 10^{-19}\ \text{J}$$

$$\nu = \frac{\Delta E}{h} = \frac{1.60 \times 10^{-19}\ \text{J}}{6.626 \times 10^{-34}\ \text{J s}} = 2.41 \times 10^{14}\ \text{s}^{-1}$$

Exp.

$\nu = 6.03 \times 10^{14}\ \text{s}^{-1}\ (\lambda = 497\ \text{nm})$

Two minutes break!



3-D one-particle Schrödinger equation

$$H = T + V = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V(x,y,z) \qquad C.M.$$

$$\hat{H} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) + V(x,y,z) \qquad Q.M.$$

$$= -\frac{\hbar^2}{2m}\nabla^2 + V(x,y,z) \longrightarrow \boxed{-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi}$$

Laplacian Operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Probability:

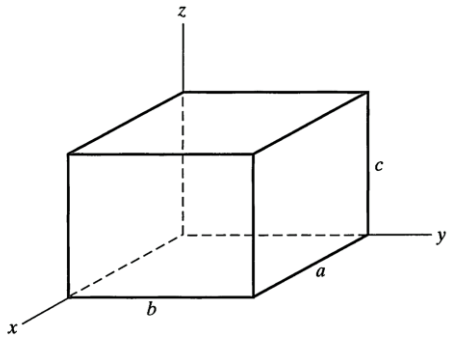
$$\left|\psi(x,y,z)\right|^2 dxdydz$$

Normalization condition:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left|\psi(x,y,z)\right|^2 dxdydz = 1$$

The particle in a 3-D box

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right)=E\psi$$



$V(x,y,z)=0$ $\quad 0<x<a,\quad 0<y<b,\quad 0<z<c$

$V(x,y,z)=\infty$ $\quad$ *elsewhere*

Separation of variables

$$\psi(x,y,z)=f(x)g(y)h(z) \qquad E_x+E_y+E_z=E$$

$$\psi(x,y,z)=\left(\frac{8}{abc}\right)^{1/2}\sin\left(\frac{n_x\pi x}{a}\right)\sin\left(\frac{n_y\pi y}{b}\right)\sin\left(\frac{n_z\pi z}{c}\right)$$

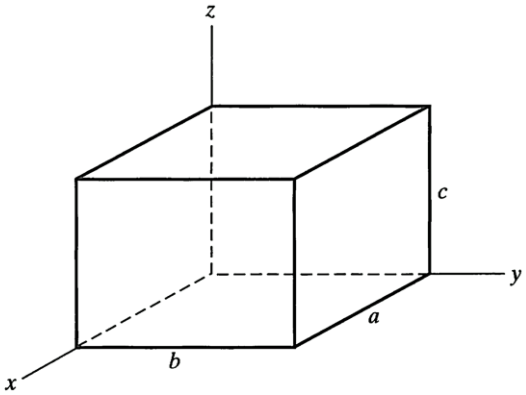
$$E=\frac{h^2}{8m}\left(\frac{n_x^2}{a^2}+\frac{n_y^2}{b^2}+\frac{n_z^2}{c^2}\right) \qquad n_x,\ n_y,\ n_z=1,\ 2,\ 3\ldots$$

A particle in a 3-D box

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right)=E\psi$$

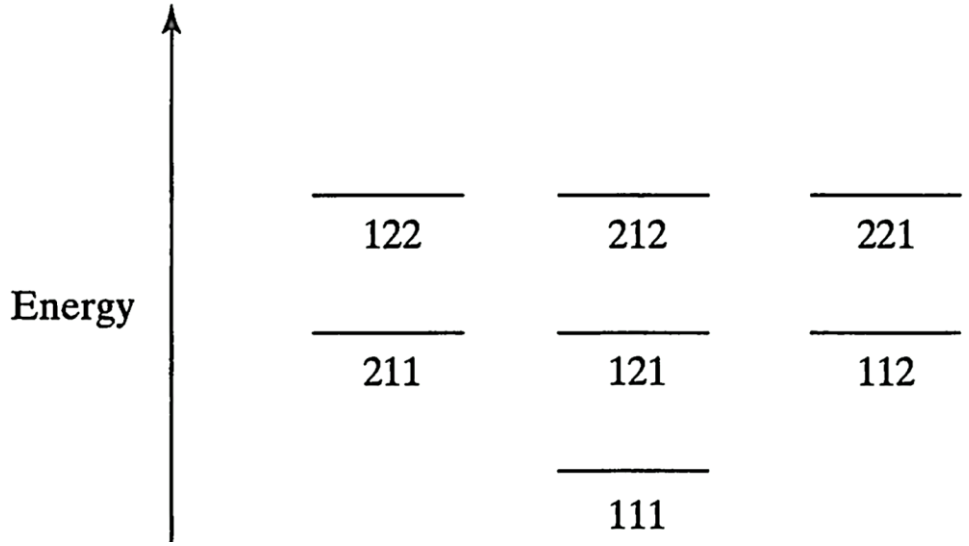
$$\psi(x,y,z)=\left(\frac{8}{abc}\right)^{1/2}\sin\left(\frac{n_x\pi x}{a}\right)\sin\left(\frac{n_y\pi y}{b}\right)\sin\left(\frac{n_z\pi z}{c}\right)$$

$$E=\frac{h^2}{8m}\left(\frac{n_x^2}{a^2}+\frac{n_y^2}{b^2}+\frac{n_z^2}{c^2}\right) \qquad n_x, n_y, n_z = 1, 2, 3\ldots$$



→ $a=b=c,$

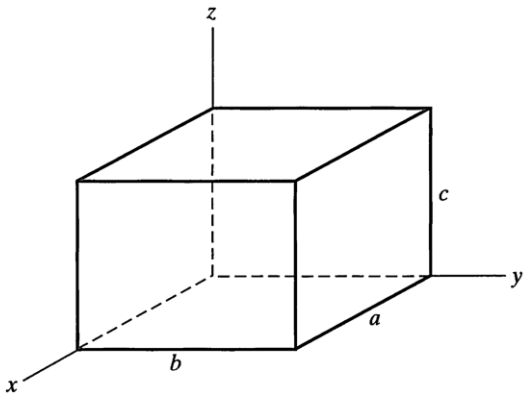
$$E=\frac{h^2}{8ma^2}\left(n_x^2+n_y^2+n_z^2\right)$$



$n_xn_yn_z$	111	211	121	112	122	212	221	113	131	311	222
$E(h^2/8ma^2)$	3	6	6	6	9	9	9	11	11	11	12

$a=b=c$,

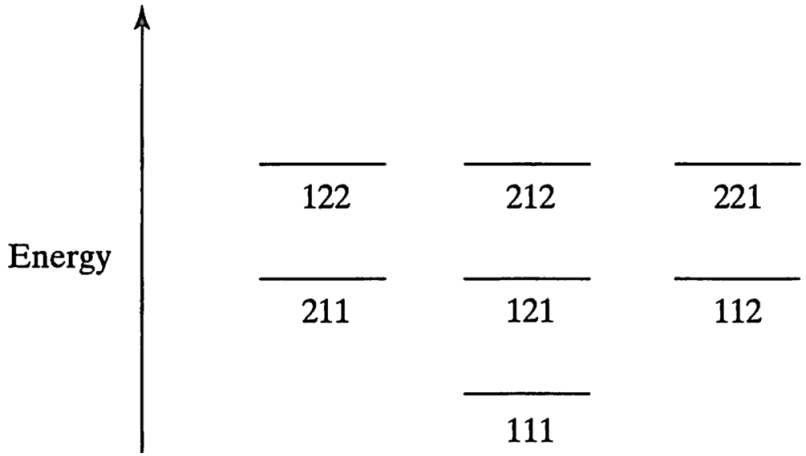
$$E = \frac{h^2}{8ma^2}(n_x^2 + n_y^2 + n_z^2)$$



Two or more independent wave functions correspond to states with the same energy eigenvalue

→ **Degenerate (简并)**

The **degree of degeneracy (简并度)** of an energy level is the number of states that have that energy.



$n_x n_y n_z$	111	211	121	112	122	212	221	113	131	311	222
$E(h^2/8ma^2)$	3	6	6	6	9	9	9	11	11	11	12

Degeneracy

A case with *n*-fold degenerate energy levels:

$$\hat{H}\psi_1 = w\psi_1, \quad \hat{H}\psi_2 = w\psi_2, \cdots, \quad \hat{H}\psi_n = w\psi_n$$

$\psi_1, \psi_2, \ldots, \psi_n$, are linearly independent

→ **Linear combination** $\phi \equiv c_1\psi_1 + c_2\psi_2 + \cdots + c_n\psi_n$

is an eigenfunction of the Hamiltonian with eigenvalue *w*

→ $\hat{H}\phi = w\phi$

Degeneracy

A case with *n*-fold degenerate energy levels:

$$\hat{H}\psi_1 = w\psi_1, \quad \hat{H}\psi_2 = w\psi_2, \cdots, \quad \hat{H}\psi_n = w\psi_n$$

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is an eigenfunction of the Hamiltonian with eigenvalue *w*

→ $\hat{H}\phi = w\phi$

$$\begin{aligned}\hat{H}(c_1\psi_1 + c_2\psi_2 + \cdots + c_n\psi_n) &= \hat{H}(c_1\psi_1) + \hat{H}(c_2\psi_2) + \cdots + \hat{H}(c_n\psi_n) \\ &= c_1\hat{H}\psi_1 + c_2\hat{H}\psi_2 + \cdots + c_n\hat{H}\psi_n \\ &= c_1 w\psi_1 + c_2 w\psi_2 + \cdots + c_n w\psi_n \\ &= w(c_1\psi_1 + c_2\psi_2 + \cdots + c_n\psi_n) \\ &= w\phi\end{aligned}$$

3-D many-particle Schrödinger equation

$$T=\frac{1}{2m_1}(p_{x_1}^2+p_{y_1}^2+p_{z_1}^2)+\frac{1}{2m_1}(p_{x_1}^2+p_{y_2}^2+p_{z_2}^2)+\cdots+\frac{1}{2m_n}(p_{x_n}^2+p_{y_n}^2+p_{z_n}^2)$$ ***C.M.***

$$\hat{T}=-\sum_{i=1}^{n}\frac{\hbar^2}{2m_i}\nabla_i^2 \qquad \nabla_i^2=\frac{\partial^2}{\partial x_i^2}+\frac{\partial^2}{\partial y_i^2}+\frac{\partial^2}{\partial z_i^2}$$ ***Q.M.***

$$V=V(x_1,y_1,z_1,\cdots,x_n,y_n,z_n)$$

$$[-\sum_{i=1}^{n}\frac{\hbar^2}{2m_i}\nabla_i^2+V(x_1,\cdots,z_n)]\psi=E\psi$$

$$\psi=\psi(x_1,y_1,z_1,\cdots,x_n,y_n,z_n)$$

Probability:

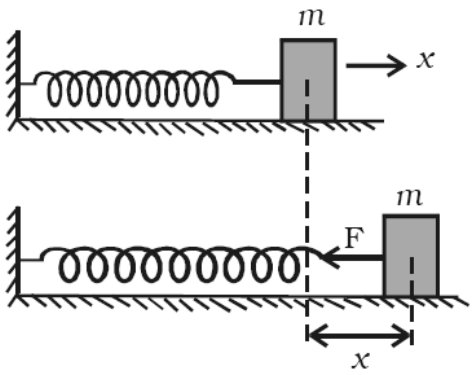
$$\left|\psi(x_1^{'},y_1^{'},z_1^{'},x_2^{'},y_2^{'},z_2^{'},\cdots,x_n^{'},y_n^{'},z_n^{'})\right|^2 dx_1dy_1dz_1dx_2dy_2dz_2\cdots dx_ndy_ndz_n$$

Five minutes break!



The 1-D Harmonic Oscillator

A good model for molecular vibrations



Classical-Mechanical Treatment

$$F = -kx = m\frac{d^2x}{dt^2} \longrightarrow x = A\sin(2\pi\nu t + b)$$

$$F = -kx$$

Force constant

Vibration frequency $\nu = \frac{1}{2\pi}\left(\frac{k}{m}\right)^{1/2}$

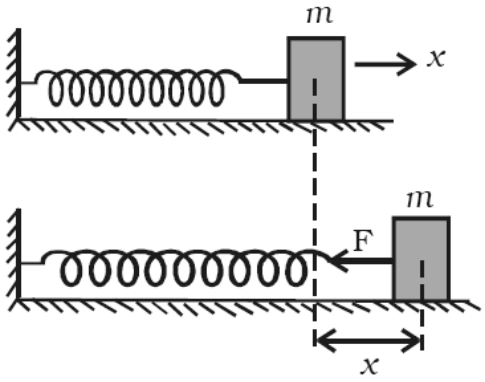
$$F = -\frac{dV}{dx} = -kx \longrightarrow V = \frac{1}{2}kx^2 = 2\pi^2\nu^2 mx^2$$

$$T = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 \longrightarrow E = T + V = \frac{1}{2}kA^2 = 2\pi^2\nu^2 mA^2$$

Turning points for the motion: $x = A$ and $-A$

The 1-D Harmonic Oscillator

Quantum-Mechanical Treatment



$$T = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 \quad , V = \frac{1}{2}kx^2 = 2\pi^2\nu^2 mx^2 \quad , \qquad \nu = \frac{1}{2\pi}\left(\frac{k}{m}\right)^{1/2}$$

$$\hat{H} = \hat{T} + \hat{V} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + 2\pi^2\nu^2 mx^2 = -\frac{\hbar^2}{2m}\left(\frac{d^2}{dx^2} - \alpha^2 x^2\right) \quad , \qquad \alpha \equiv \frac{2\pi\nu m}{\hbar}$$

$$\frac{d^2\psi}{dx^2} + (2mE\hbar^{-2} - \alpha^2 x^2)\psi = 0$$

The general solution

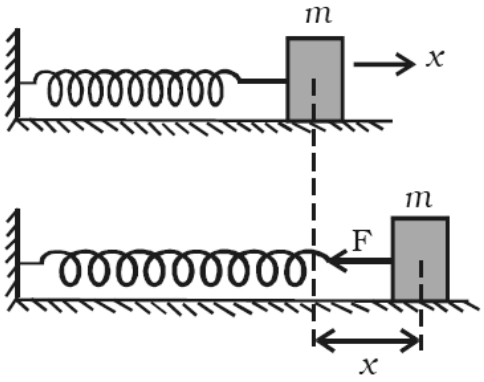
$$\psi = Ae^{-\alpha x^2/2}\sum_{l=0}^{\infty} c_{2l+1}x^{2l+1} + Be^{-\alpha x^2/2}\sum_{l=0}^{\infty} c_{2l}x^{2l}$$

$$E_\upsilon = (\upsilon + \frac{1}{2})h\nu, \quad \upsilon = 0,1,2,\cdots$$ ← **Quantum number**

$$E_\upsilon = (\upsilon + \frac{1}{2})\hbar\omega, \quad \omega = \left(\frac{k}{m}\right)^{1/2}$$ ← **Angular frequency**

The 1-D Harmonic Oscillator

Quantum-Mechanical Treatment



$$\frac{d^2\psi}{dx^2} + (2mE\hbar^{-2} - \alpha^2 x^2)\psi = 0 \qquad \alpha \equiv \frac{2\pi\nu m}{\hbar}$$

$$\psi = Ae^{-\alpha x^2/2}\sum_{l=0}^{\infty} c_{2l+1}x^{2l+1} + Be^{-\alpha x^2/2}\sum_{l=0}^{\infty} c_{2l}x^{2l}$$

$$E_\upsilon = (\upsilon + \frac{1}{2})h\nu, \quad \upsilon = 0,1,2,\cdots$$

$$E_\upsilon = (\upsilon + \frac{1}{2})\hbar\omega, \quad \omega = \left(\frac{k}{m}\right)^{1/2}$$

$$\longrightarrow \quad E_0 = \frac{1}{2}\hbar\omega$$

Zero-point energy

